

Chickpea (Gram)

Cicer arietinum — Pakistan's Leading Pulse Crop

A Complete Cultivation Guide for

Punjab · Sindh · Khyber Pakhtunkhwa · Balochistan

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1. Introduction

Chickpea, known locally as chana or gram, is Pakistan's most important pulse (legume) crop and a key source of plant protein for millions of households. It is a hardy, mostly rainfed crop grown in the drier parts of the country, especially in Punjab's Thal and Pothohar regions and large tracts of Balochistan and Sindh, where water for irrigated crops is limited. Chickpea is valued not just for food (as whole gram, split dal, or roasted snacks) but also for improving soil fertility, since like other legumes it can fix atmospheric nitrogen through root nodules. This guide covers how chickpea is grown across Pakistan's main provinces and what practices help farmers achieve better and more reliable yields.

2. Scientific Name

Chickpea's scientific name is *Cicer arietinum*, a member of the Fabaceae (legume) family. Pakistan mainly grows the desi type (small, dark, angular seeds) though some kabuli type (larger, cream-colored, smooth seeds) is also cultivated, especially for export and specialty markets.

3. Importance in Pakistan

Chickpea covers close to one million hectares in an average year, making it by far the largest pulse crop in the country and a major contributor to the total pulses supply. It is central to Pakistani diets as a cheap, protein-rich food, used as whole boiled gram (choley), ground into besan flour for numerous snacks and dishes, and roasted as a popular snack food. Because Pakistan does not produce enough pulses to fully meet domestic demand, chickpea imports (along with other pulses) add to the food import bill, so improving domestic chickpea yields has direct implications for both nutrition security and the balance of trade. Its ability to grow on marginal, low-rainfall land with minimal irrigation also makes it a vital crop for resource-poor farmers in Balochistan, southern Punjab, and parts of Sindh.

4. Major Producing Provinces and Districts

- Punjab: Bhakkar, Layyah, Mianwali, Khushab, Dera Ghazi Khan, and the Thal desert tract — the country's largest chickpea-growing region, mostly rainfed (barani).
- Sindh: Thar Parkar, Sanghar, and parts of upper Sindh — grown on residual soil moisture after monsoon rains.
- Balochistan: Loralai, Zhob, Qilla Saifullah, Pishin — significant rainfed and some irrigated area, valued for good grain quality.
- Khyber Pakhtunkhwa: D.I. Khan, Bannu, Kohat — grown on both rainfed uplands and some irrigated plains.

5. Climate Requirements

Chickpea is a cool-season crop that grows well with average temperatures of 15–25°C during vegetative growth, and it needs cooler nights for good pod set. It is relatively drought-tolerant once established, which suits Pakistan's rainfed tracts, but too much rain or humidity during flowering and pod-filling encourages the serious fungal disease *Ascochyta* blight. Frost during flowering can cause significant pod and flower

loss, especially in the higher, cooler parts of Balochistan and northern Punjab.

6. Soil Requirements

Chickpea grows best in well-drained sandy loam to clay loam soils with a near-neutral pH (6.0–7.5), and it tolerates moderately alkaline soils common in Punjab and Sindh better than many other pulses. Good drainage is essential, since chickpea is very sensitive to waterlogging, which quickly causes root rot. Heavy, poorly drained clay soils should be avoided or well-leveled and drained before sowing.

7. Land Preparation

Because chickpea is mostly rainfed, land preparation focuses on conserving soil moisture: one or two ploughings followed by planking to break clods and create a fine, firm seedbed that reduces moisture loss. In areas relying on residual moisture after monsoon rains, minimum tillage just ahead of sowing is often preferred to avoid drying out the topsoil. Fields should be reasonably leveled to prevent waterlogging in low pockets after occasional rains.

8. Recommended Varieties (Pakistan)

- Punjab: Noor 2009, Bittal 2016, Thal 2006, and Karak 3 (also grown in KP) are widely recommended desi varieties known for good yield and wilt tolerance.
- Balochistan and rainfed areas: Varieties bred for drought tolerance and short duration, developed and tested by the Ayub Agricultural Research Institute (AARI) and the Pulses Research Institute.
- Kabuli type: CM 2008 and similar large-seeded varieties are grown in limited areas for premium and export markets.
- Farmers are advised to use the latest varieties recommended each season by the Punjab Seed Corporation, the Barani Agricultural Research Institute (BARI, Chakwal), and provincial extension departments, since new blight- and wilt-resistant releases regularly replace older varieties.

9. Seed Rate

Recommended seed rate is about 25–30 kg per acre for desi types under normal rainfed conditions, and slightly higher (35–40 kg per acre) for kabuli types due to their larger seed size. Seed rate may be increased somewhat under poor moisture conditions or late sowing to compensate for expected lower germination and establishment.

10. Seed Treatment

Seed should be treated with a fungicide such as thiram or carbendazim (2–3 g per kg seed) to protect against seed-borne fungal diseases and seedling blight. Because chickpea benefits greatly from nitrogen-fixing bacteria, seed should also be inoculated with the appropriate *Rhizobium* (*Mesorhizobium*) culture just before sowing wherever the crop has not been grown recently in that field, which can meaningfully boost yield without extra nitrogen fertilizer cost.

11. Sowing Time (Province-wise)

- Punjab (rainfed Thal/Pothohar): Sown mid-October to mid-November, taking advantage of post-monsoon soil moisture.
- Punjab (irrigated areas): Sown late October to November.
- Sindh: Sown October–November, mainly on residual moisture in Thar Parkar and upper Sindh.
- Balochistan: Sown October–November in most areas; slightly later in cooler, higher elevations.
- Khyber Pakhtunkhwa: Sown October–November in rainfed uplands, aligned with the onset of cooler weather.

12. Sowing Methods

Chickpea is generally sown by drilling seed in rows 30 cm apart at a depth of 8–10 cm to reach adequate soil moisture, using a seed drill where available for uniform depth and spacing, which is important for good germination in dry rainfed conditions. Broadcasting followed by ploughing is still practiced by some smallholders but generally gives poorer stand establishment and more uneven crop growth than row sowing.

13. Fertilizer Recommendations (NPK)

Because chickpea fixes much of its own nitrogen through Rhizobium bacteria, fertilizer needs are modest: a general recommendation is 20–25 kg Nitrogen (mainly as a small starter dose), 60 kg Phosphorus (P₂O₅), and, where soils are deficient, 30 kg Potash (K₂O) per acre, applied fully at sowing. Phosphorus is especially important for good root and nodule development in chickpea, so adequate phosphorus application, guided by soil testing, generally gives a better yield response than extra nitrogen.

14. Irrigation Schedule

Most of Pakistan's chickpea crop is grown as a rainfed (barani) crop relying on residual soil moisture and winter rains, with no irrigation applied at all in many areas of Punjab's Thal tract and Balochistan. Where irrigation is available, one or two supplemental irrigations at the pre-flowering and pod-filling stages can significantly boost yield, especially in dry years, but care should be taken to avoid over-irrigation, which increases the risk of root rot and Ascochyta blight.

15. Weed Management

Chickpea grows relatively slowly in its early stages and can be outcompeted by weeds if fields are not kept clean, so one to two hand weedings or hoeings around three to six weeks after sowing are recommended. Pre-emergence herbicides approved for chickpea are used on some larger farms to reduce labor costs, particularly where labor is scarce during the sowing season.

16. Insect Pests

The most damaging insect pest of chickpea in Pakistan is gram pod borer (*Helicoverpa armigera*), whose caterpillars bore into developing pods and destroy seeds; this pest also attacks cotton and other crops, so regional cropping patterns affect its buildup. Cutworms can damage young seedlings, and aphids may transmit viral diseases. Management includes regular field scouting, pheromone traps, encouraging natural predators, and timely, need-based insecticide application when pod borer populations exceed threshold levels.

17. Diseases

Ascochyta blight is the most serious disease of chickpea in Pakistan, especially damaging in years with wet, cool weather during flowering, capable of causing complete crop failure if unmanaged. Fusarium wilt is a major soil-borne disease that can persist in fields for years, making crop rotation and resistant varieties essential. Root rot and collar rot become serious under waterlogged or poorly drained conditions. Management relies primarily on resistant varieties, certified disease-free seed, crop rotation with cereals, and fungicide seed treatment, supplemented by foliar fungicide sprays during high blight risk periods.

18. Deficiency Symptoms

Nitrogen deficiency, though less common due to nitrogen fixation, shows as pale yellow-green older leaves, often linked to poor *Rhizobium* nodulation. Phosphorus deficiency causes stunted growth, delayed flowering, and poor root and nodule development. Potassium deficiency appears as yellowing and scorching along leaf margins. Zinc and iron deficiencies can occur in the alkaline, calcareous soils common in parts of Punjab and Sindh, showing as interveinal yellowing of younger leaves.

19. Harvesting

Chickpea is ready for harvest when the plants turn yellowish-brown and the leaves start to dry and shed, and pods rattle when shaken, typically 90 to 150 days after sowing depending on variety and sowing date. Harvesting is usually done by hand-pulling or cutting the plants close to the ground, followed by drying in the field for a few days before threshing, traditionally by beating or trampling, though mechanical threshers are increasingly used on larger farms.

20. Storage

Threshed chickpea grain must be dried to a safe moisture level (below about 10–12%) before storage to prevent mold and insect infestation. Grain is typically stored in clean, dry gunny bags or metal bins in well-ventilated stores, protected from moisture and pests such as pulse beetles, which can cause significant losses in stored chickpea if fumigation or other pest control measures are not applied.

21. Expected Yield in Pakistan

The national average yield of chickpea in Pakistan is relatively low, around 600–800 kg per hectare, mainly because most of the crop is grown under rainfed conditions with minimal inputs and is vulnerable to drought and disease. Under good management with resistant varieties, adequate phosphorus, *Rhizobium*

inoculation, and timely disease control, yields of 1,200–1,500 kg per hectare or more are achievable, showing substantial room for improvement even without irrigation.

22. Government Recommendations

The Pulses Research Institute at Faisalabad, the Barani Agricultural Research Institute at Chakwal, and PARC regularly test and recommend new chickpea varieties with improved blight and wilt resistance, along with agronomic packages suited to Pakistan's rainfed tracts. Government pulses development programs periodically promote certified seed distribution, Rhizobium inoculant use, and balanced fertilizer application to help close the large gap between current farmer yields and research-station yields.

23. Modern Farming Practices

Zero or minimum tillage sowing directly into standing stubble is being promoted in some rainfed areas of Punjab and KP to conserve scarce soil moisture and reduce production costs. Use of certified seed and Rhizobium inoculant, though still not universal, is spreading through extension and NGO programs as farmers see the yield benefits. Mechanical harvesters and combine threshers are gradually replacing manual threshing on larger commercial farms.

24. Precision Agriculture

Precision practices in chickpea are still limited given its predominantly rainfed, smallholder production base, but soil testing to guide phosphorus application, and satellite or weather-based advisories to help time sowing to available soil moisture, are being introduced by some research institutes and mobile advisory services. Drought forecasting information is increasingly used to help farmers decide planting dates and expected risk in marginal rainfall years.

25. Sustainable Farming

Chickpea itself is a naturally sustainable crop choice because of its nitrogen-fixing ability, low water requirement, and suitability for marginal land, making it valuable in crop rotations with wheat or other cereals to improve soil fertility and break disease cycles. Encouraging Rhizobium inoculation instead of synthetic nitrogen, promoting resistant varieties to reduce fungicide dependence, and adopting minimum tillage to conserve soil moisture and reduce erosion are key sustainability practices being promoted across Pakistan's rainfed chickpea belts.

26. Regional Notes

Chickpea remains fundamentally a rainfed crop of Punjab's Thal and Pothohar tracts, Balochistan's uplands, and parts of Sindh and KP, meaning its success each year is closely tied to the timing and amount of monsoon and winter rainfall. Improving access to resistant varieties, Rhizobium inoculant, and basic phosphorus fertilization offers the most practical path to higher, more stable yields for Pakistan's chickpea farmers, most of whom operate with limited irrigation and modest input budgets.